C. KARTHIK RAJA

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CAREER OBJECTIVE

To begin my career as an AI/ML Engineer by applying a strong foundation in Python, Machine Learning, and Deep Learning, and to contribute to innovative solutions while continuously enhancing my technical and problem solving skills.

EDUCATION

B.E. Computer Science and Engineering — Expected 2027
Nehru Institute of Technology (Autonomous) — CGPA 8.67 · through 6th semester

Higher Secondary — Class XII (CBSE) — 2022
Sri Ramakrishna Public School — 78.6%

Secondary — Class X (CBSE) — 2020
Sri Ramakrishna Public School — 89.9%

TECHNICAL SKILLS

Languages: Python 3, C
Web Development: HTML, CSS, Bootstrap, Django
AI Foundations: Machine Learning, Deep Learning, Neural Networks, Genai(Learning)
Libraries & Frameworks: NumPy, Pandas, DistilBERT
Databases & APIs: MySQL
Tools: Git, GitHub, Docker, VS Code, PyCharm, Jupyter Notebook
Soft Skills: Communication, Quick Learning, Logical Thinking

PROJECTS

News Research Tool *Python · LangChain · RAG · FAISS · Ollama · Streamlit*

- Built a RAG-based news research application that processes article URLs and enables context-aware question answering.
- Implemented document chunking, HuggingFace embeddings, FAISS similarity search, and retrieval-based context generation.
- Integrated Ollama Llama 3.2 for local LLM inference and developed a Streamlit interface with source references.

AI Mental Health Assistant *Python · TensorFlow · DistilBERT · FastAPI · React.js · MySQL · Whisper*

- Fine-tuned DistilBERT for emotion classification from text and built an end-to-end AI inference pipeline covering preprocessing, prediction, confidence estimation, and personalized CBT-inspired response generation.
- Developed a FastAPI backend with JWT authentication and MySQL integration for secure user management and persistent chat history; integrated OpenAI Whisper for voice and text interaction.
- Built a React frontend with dark/light themes, emotion analytics dashboard, wellness points, streak tracking, and progress visualization to drive user engagement.

ATM Transaction System *Python · MySQL · Twilio API*

- Developed a console-based ATM system supporting deposit, withdrawal, and balance inquiries.
- Integrated MySQL for secure transaction storage and Twilio API for real-time SMS notifications.
- Implemented user authentication and session management to ensure transaction security.

PRESENTATIONS & CONTESTS

- **1st Place** — Pen & Program coding event, TechGeeks 2025, Dhaanish Ahmed Institute of Technology, Coimbatore (8 April 2025).
- **3rd Place** — "AI Powered Rail Safety for Public Benefit," TechGeeks 2025 National Level Symposium, Dhaanish Ahmed Institute of Technology, Coimbatore (8 April 2025).
- **Presented** "AI Powered Rail Safety for Public Benefit" at Tech Magnetz 2024 National Level Technical Symposium, Hindusthan Institute of Technology, Coimbatore (26 March 2024).

CERTIFICATIONS & WORKSHOP

- **Supervised Machine Learning: Regression & Classification** — DeepLearning.AI & Stanford Online (Coursera), May 2026.
- **Python for Data Science, AI & Development** — IBM on Coursera.
- **AI Foundations Associate** — Oracle University.
- **NPTEL Certification** — Indian Institute of Technology Madras (Roll No. NOC25CS60S443407364).
- **Workshop: Green Skills and Python** — Nehru Institute of Technology; explored Python for data analysis, climate research, and eco-friendly system modeling.

EXTRA-CURRICULAR ACTIVITIES

- **Student Council President** — Nehru Institute of Technology (2026–2027). Elected to lead the Student Council, coordinate student activities, and represent student interests while collaborating with college administration.
- Active participant in national-level technical symposia and inter-college coding competitions.
- Engaged in collaborative academic projects, demonstrating teamwork and communication skills.
- Continuously upskilling through Coursera, NPTEL, and Oracle University.

PERSONAL DETAILS

Date of Birth: 04 April 2006
Father's Name: A. Chinnadurai
Languages Known: Tamil, English
Address: 8/174, West Street, T. Kudikadu, Thathanur Post, Udayarpalayam, Ariyalur

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief. I take full responsibility for the authenticity of all details mentioned in this resume.

Place:

Date:

C. Karthik Raja